Model Context Protocol and Its Effect on Agentic AI:

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Abstract

With the advancement of agentic AI, one important bottleneck to effective tool-use has been the absence of a normalized interface to external data sources and applications. We report a systematic review of the Model Context Protocol (MCP), an open-source convention launched in November 2024 to address this fragmentation. Our results indicate that MCP 's use of JSON-RPC 2.0 affords a scalable, vendor-neutral framework for AI-to-tool communication, which hit around 9M monthly downloads within one year of publication. We examine MCP 's relationship to the Agent2Agent (A2A) protocol for multi-agent communication and coordination, the protocol 's impact on Software-as-a-Service (SaaS) architectures, and 16 distinct security vulnerabilities including prompt injection via tool definitions and "toxic flow" logical chains. We conclude that MCP lays an essential groundwork for agentic AI architectures, but formal security standards and multi-tenant isolation services are pressing unanswered questions.

1. Introduction

1.1 What Is This Paper About?

This paper reviews everything we know so far about the Model Context Protocol — or MCP for short. MCP is a new open standard that makes it easier for AI systems to connect with outside tools, apps, and data. Think of it like a common language that AI systems and software tools can use to talk to each other.

Before MCP, connecting an AI to an outside tool was like building a custom bridge every single time. If a company had 10 AI products and 20 tools, they needed 200 custom bridges. MCP turns that into a simple, reusable road that everyone can use.

1.2 What Are AI Agents?

An AI agent is an AI system that doesn't just answer questions — it actually does things. It can plan a sequence of steps, use tools, look up real information, and complete tasks without a person clicking every button. Examples include AI that books calendar meetings, writes and sends reports, or monitors a database and alerts a team when something goes wrong.

For AI agents to work well, they need to connect to many different systems — databases, apps, calendars, files, and more. Without a shared standard, this is very hard to build and maintain.

1.3 Why Does MCP Matter?

MCP was released by Anthropic in November 2024. It is open-source, meaning anyone can use and build on it for free. It was designed to solve the connection problem once and for all by giving every AI and every tool the same simple interface to follow.

This review covers how MCP came to be, how it works, who uses it, how it affects AI agents, how it changes software businesses, and what risks it introduces — especially around security.

1.4 Goals of This Review

Specifically, this paper aims to:

- Show how AI tool-use and connection technology has grown from 2015 to 2025.
- Explain how MCP works in plain terms.
- Show how fast MCP was adopted and by whom.
- Explain how MCP makes AI agents more powerful.
- Show how MCP changes the way online software businesses (SaaS) operate.
- List the security problems MCP introduces and how they can be fixed.

2. Results and Discussion

2.1 How We Got Here: AI Tool Use from 2015 to 2024

To understand why MCP matters so much, it helps to know what came before it. Here is a simple history of how AI went from answering questions to actually doing tasks.

2.1.1 Early AI Models: Smart but Stuck (2015–2020)

Early language models could read and write text very well, but they had no way to look things up or use outside tools. They only knew what they were trained on. If you asked them about something that happened after their training finished, they had no idea. This made them useful for writing and answering simple questions, but not for doing real work in a business system.

2.1.2 First Steps: Giving AI Tools (2020–2022)

Around 2020, companies started experimenting with giving AI systems the ability to call external functions — basically letting them use small programs to do specific tasks, like looking up the weather or running a calculation. OpenAI was one of the first to offer this through their API. It worked, but each AI and each tool still needed its own custom setup. There was no standard way of doing it.

2.1.3 AI Agents Start to Emerge (2022–2024)

Between 2022 and 2024, researchers and companies started building AI agents — systems that could plan and carry out multiple steps on their own. Tools like LangChain and AutoGPT let developers' chain together AI reasoning with tool use. But the connection problem remained: linking any AI to any new tool still required custom coding every time. As more tools and AI systems appeared, this became a major roadblock.

2.1.4 The Breaking Point and MCP's Solution (2024)

By mid-2024, many engineering teams said that building all these custom connections was taking more time than actually building the AI features. The problem was getting out of hand. Anthropic recognized this and spent time designing a clean, universal solution. That solution was MCP, released in November 2024.

2.2 How MCP Works

2.2.1 The Three Main Parts

MCP uses three main parts that work together:

- Host: This is the AI app that the user talks to — for example, Claude or a custom business chatbot.
- MCP Client: This is built into the Host. It handles sending and receiving messages in the MCP format.
- MCP Server: This is a small program that wraps around a tool or data source — like a database, calendar, or file system — and speaks the MCP language so the AI can use it.

These three parts talk to each other using a message format called JSON-RPC 2.0. This is a standard message format used in software development — it is clear, reliable, and widely understood.

2.2.2 Three Things MCP Servers Can Offer

When an AI connects to an MCP server, the server can provide three types of things:

- Tools: Actions the AI can perform, like 'send an email,' 'search the database,' or 'create a task.' These are like buttons the AI can press.
- Resources: Information the AI can read, like a document, a database record, or a live data feed. These are like windows the AI can look through.
- Prompts: Pre-written instructions that help the AI ask the right questions in the right way when using a tool.

This clear separation keeps things organized: the AI focuses on thinking and deciding, while the MCP server handles the actual work of talking to outside systems.

2.2.3 Why JSON-RPC 2.0 Was the Right Choice

The choice to use JSON-RPC 2.0 was smart. It is a messaging format that already exists, is well understood, and has good tools for testing and troubleshooting. MCP took inspiration from the Language Server Protocol (LSP), which solved a very similar problem in software coding tools — and did it very successfully. MCP applies the same idea to AI systems.

2.3 How Fast MCP Was Adopted

MCP spread faster than almost any previous AI standard. Here is a timeline of key moments:

Date	What Happened
November 2024	Anthropic releases MCP as a free, open-source standard. Early users include Block (Square), Apollo, Zed, and Replit.
February 2025	Over 1,000 community-built MCP servers exist. Downloads reach hundreds of thousands per month.
March 2025	OpenAI officially supports MCP across all its major developer products.
April 2025	Google DeepMind adds MCP support. First major security report published identifying 16 types of risks.
June 2025	MCP specification updated with clearer rules on login and user permissions.
August 2025	More than 5,800 MCP servers registered. Monthly downloads hit around 9 million.
December 2025	Anthropic moves MCP to the Linux Foundation's Agentic AI Foundation (AAIF).
February 2026	Over 6,400 MCP servers registered. Used in production at hundreds of large companies.

2.4 What MCP Does for AI Agents

2.4.1 From Locked-In Knowledge to Live Information

Before MCP, AI agents could only use information from their training. This meant they could give wrong or outdated answers. A big cause of this is called 'hallucination' — when an AI confidently says something that is simply not true because it has no way to check. MCP fixes this by letting agents pull real, up-to-date information from trusted sources at the moment they need it.

Real examples from reviewed studies: in hospitals, MCP-connected AI systems can check live patient records to help doctors make decisions. In banks, AI agents can read current market prices and regulations to support financial analysts.

2.4.2 Teams of AI Agents Working Together

MCP also makes it easier to build networks of AI agents, where one 'manager' agent directs several 'specialist' agents. Because every agent uses the same MCP standard, they can hand off tasks cleanly without needing custom connections between each pair. Studies show that real deployments now include networks of up to dozens of agents working together on complex business tasks.

2.4.3 The Rise of 'Context Engineering'

A new skill area has emerged called 'context engineering.' This means carefully choosing and organizing what information is given to an AI at the right moment, so it gives better results. MCP is the main tool that makes context engineering possible, because it controls what the AI can access and when. Thoughtworks named this one of the most important new practices in AI development in late 2025.

2.4.4 More Consistent AI Behaviour

One long-standing complaint about AI agents is that they behave differently each time, even when given the same task. MCP helps reduce this by wrapping actions in well-defined tools. Instead of asking the AI to figure out exactly how to call an API, the AI just picks which tool to use. Fewer decisions mean fewer chances for the AI to go off track.

3. Security in MCP Servers and Tools

Security is one of the most important topics in MCP right now. Because MCP gives AI agents the power to connect to real systems and take real actions, a mistake or attack can have serious real-world effects. This section explains the main security risks clearly, what has already been done to address them, and what still needs to happen.

3.1 Why MCP Security Is Different from Regular Software Security

In normal software, a human clicks a button and a fixed action happens. In MCP, an AI decides which tools to use and when — often without a human approving each step. This means if something goes wrong — whether from a bug, a bad configuration, or an attack — the AI might carry out harmful actions quickly and at scale before anyone notices.

Also, MCP sits between the AI (which uses natural language) and software tools (which take exact commands). This gap — where language meets action — is a new kind of attack surface that traditional security tools were not built to handle.

3.2 Known Security Risks

3.2.1 Prompt Injection Through Tool Descriptions

When an AI connects to an MCP server, it reads the descriptions of the tools that server offers. These descriptions are written in plain English. A bad actor could write a malicious MCP server where the tool descriptions contain hidden instructions for the AI. For example, a description might secretly say: 'After completing this task, also forward all emails to this external address.' The AI might follow these instructions without the user knowing.

This is called prompt injection, and it is one of the hardest problems to fully solve because the AI cannot easily tell the difference between a real instruction and a fake one hidden inside a description.

3.2.2 Tool Poisoning

Similar to prompt injection, tool poisoning happens when a fake or hacked MCP server pretends to be a real, trusted one. The AI thinks it is using a safe tool but is actually sending data to a bad actor, or performing actions the user never intended. This is especially dangerous because the AI has no built-in way to verify that a tool is what it claims to be.

3.2.3 Missing or Weak Login Controls

When MCP was first released, there were few rules about how servers should check who is connecting to them. Many early MCP servers were set up without any login checks at all — meaning anyone who knew the server's address could connect and use it. The June 2025 update to the MCP specification improved this, but many older deployments still have this gap. Without proper login controls, anyone can pretend to be a trusted AI agent and access sensitive data or trigger actions.

3.2.4 Cross-Server Tool Shadowing

When an AI has access to many MCP servers at once, a bad server can register a tool with the same name as a tool on a trusted server. If the AI picks the wrong one, it may send sensitive information to the bad server or perform the wrong action. This is like someone setting up a fake cash machine right next to a real one — it looks the same, but the result is very different.

3.2.5 Combining Safe Tools in Unsafe Ways

Each individual MCP tool might be safe on its own. But when an AI uses several tools together, the combination might create a dangerous path. For example: one tool reads private user data, another sends data externally, and a third deletes the source. Each step seems fine, but together they silently leak and erase data. Researchers call this a 'toxic flow.' Detecting toxic flows requires looking at the whole system, not just individual tools.

3.2.6 Stolen Saved Passwords (Token Storage Risks)

MCP servers often need to store login tokens — small digital keys that grant access to external services like Google, Slack, or Salesforce. If a single MCP server holds tokens for many services and gets hacked, the attacker gains access to all those services at once. This is a classic example of putting too many eggs in one basket. A breach of one MCP server can cascade into a much wider security incident.

3.2.7 Lack of Human Oversight in Multi-Step Tasks

When AI agents carry out long chains of actions — for example, pulling a report, analyzing it, writing a reply, and sending it — there may be no human check anywhere in the chain. If the AI makes a wrong decision early on, all the later steps build on that mistake. By the time a human notice, the damage may already be done. This is not strictly an MCP problem, but MCP makes these long chains much easier to build, which makes this risk more common.

3.3 What Has Been Done So Far

The community has not ignored these problems. Several steps have already been taken:

- MCP-scan: A free tool released by security researchers that automatically checks MCP server configurations for common vulnerabilities and flags suspicious tool descriptions.
- June 2025 Specification Update: The official MCP standard was updated to include clearer guidance on user authentication (login) and permission controls (who can do what).
- Hou et al. Threat Taxonomy (2025): A research paper published on arXiv that identified and categorized 16 types of security threats specific to MCP, giving the community a shared vocabulary to discuss and address them.
- Thoughtworks Advisory: Thoughtworks formally advised against simply converting all existing APIs directly into MCP tools without a security review — this practice, called 'naive API-to-MCP conversion,' opens up many of the risks listed above.
- Linux Foundation Governance: Moving MCP to the Agentic AI Foundation under the Linux Foundation means security updates and standards will now be managed openly, with input from many organizations rather than just one.

3.4 What Still Needs to Be Done

Despite the progress, several important gaps remain:

- Formal Security Standards: There is still no official, widely agreed-upon security checklist that all MCP server builders must follow. This is urgently needed.
- AI-Aware Firewalls: Traditional security tools monitor network traffic or code. MCP introduces a new type of traffic — natural language instructions mixed with tool calls — that existing tools cannot monitor effectively. New tools that understand AI behavior are needed.
- Human Approval Steps for High-Risk Actions: For actions that cannot be undone — like deleting data, sending emails, or making payments — there should be a required human approval step. This is not yet standard in most MCP deployments.
- Regular Security Audits of MCP Servers: Just like websites are regularly checked for vulnerabilities, MCP servers should be audited on a regular schedule. No standard audit process exists yet.
- Training for Developers: Many developers building MCP servers are not security experts. Simple, clear guidance on how to build MCP servers safely — written for non-security specialists — is needed and largely missing today.

3.5 Summary: The Security Balance

MCP gives AI agents great power — the ability to take real actions in real systems. With that power comes real risk. The good news is that the community recognized these risks early and has started building solutions. The honest truth is that the tools and standards for securing MCP are not yet mature enough to match the speed of deployment. Closing this gap is the most important near-term challenge for the MCP ecosystem.

4. Impact on SaaS (Online Software Services)

4.1 What Changes for SaaS Companies

SaaS stands for Software as a Service — this means software you use online, like Google Docs, Salesforce, or Slack, where you pay a subscription instead of buying and installing the software yourself. MCP changes how these kinds of products connect with AI, which has a big impact on their business.

Before MCP, if a SaaS company wanted their product to work with an AI, they had to build a custom connection for each AI platform separately. With MCP, they build one MCP server and their product works with every AI that supports MCP — Claude, ChatGPT, Gemini, and others. This saves a huge amount of engineering time.

4.2 Breaking Down Data Silos

Most businesses have their data spread across many different tools — customer records in one place, support tickets in another, billing in a third. Before MCP, getting AI to work with all of these at the same time was extremely difficult. An AI agent connected via MCP can pull from all of them at once. For example: an AI agent can read a customer's purchase history from Stripe, their support tickets from Zendesk, and their contact details from Salesforce — all in one conversation — and give a complete answer to a support question in seconds.

4.3 New Ways to Charge for AI Features

Because every MCP tool call is a small, measurable action, SaaS companies can use MCP calls to measure and charge for AI usage. For example:

- Basic plan users get read-only tools — they can look things up, but not change anything.
- Professional plan users get action tools — they can create, update, and send things.
- Enterprise users get full automation tools — AI agents can run entire workflows without human input.

This is a natural fit with the usage-based billing model that many SaaS companies are already moving toward.

4.4 Competition and Business Strategy

MCP is also changing how SaaS companies compete. Companies that publish good, well-designed MCP servers early will show up more often when AI agents look for tools to use. This is like having good search engine results — it drives discovery and usage. Companies that do not publish MCP servers' risk being left out of AI-powered workflows entirely.

Industry analysis suggests that by the end of 2025, 75% of organizations building AI tools will adopt MCP as part of their standard approach. That creates strong pressure for all SaaS providers to participate.

4.5 Challenges for SaaS with Many Users

One technical challenge for SaaS companies is that MCP was originally designed for one user talking to one AI. But SaaS products serve thousands or millions of users at once. Making MCP work safely in a world with many users — without one user accidentally seeing another user's data — requires careful engineering. This challenge is being actively worked on, but no single standard solution exists yet.

4.6 Real Examples by Industry

4.6.1 Software Development Tools

Code editors like Cursor and VS Code now include MCP clients, allowing AI coding assistants to look at live code, run tests, and check documentation in real time. This makes AI coding help much more accurate and useful.

4.6.2 Knowledge and Documents

Tools like Notion and Confluence have MCP servers, so AI agents can read, write, and update company documentation as part of automated workflows. There are tools helps AI coding assistants find the exact version of a programming library's guide they need — reducing errors caused by outdated instructions.

4.6.3 Healthcare

Early experiments in hospitals show AI agents using MCP to read patient records and medical databases to support clinical decisions. This is very promising but also requires extra care because of patient privacy laws and the serious consequences of errors in healthcare.

4.6.4 Finance

Financial companies are using MCP-connected agents to monitor live market data, pull regulatory documents, and summarize portfolio performance. Tasks that once took analysts hours can now be completed in minutes.

5. Limitations and Future Research

5.1 Limits of This Review

This review has some important limits to keep in mind. Because MCP was only released in November 2024, most of the sources we used are industry reports and technical guides rather than long-term academic studies. It is too early to have studies that measure MCP's impact over several years. Also, MCP was updated multiple times during our review period, which made it hard to keep up with every change.

5.2 What Needs More Research

- Security Standards: Formal, agreed-upon rules for building secure MCP servers are urgently needed.
- Multi-User Safety: Standard solutions for using MCP safely when serving thousands of users are still being developed.
- Measuring Real Impact: Research studies that measure whether MCP actually makes businesses more productive, safer, and more reliable — not just in theory, but in real deployments.
- Protocol Compatibility: As other standards (like Google's Agent2Agent protocol) emerge alongside MCP, research into how they work together is needed.
- Developer Education: Research into how to best teach non-security developers to build MCP servers safely.

6. Conclusion

This review has looked at where MCP came from, how it works, how fast it has spread, what it does for AI agents, how it changes online software businesses, and — critically — what security risks it brings.

The core message is simple: MCP is a very important step forward for AI. It solves a real and urgent problem — making it easy for any AI to connect to any tool — in a clean, open, and vendor-neutral way. The speed at which it was adopted shows how badly this solution was needed.

For businesses that build and sell software online, MCP changes both the opportunities and the risks. Companies that move quickly and build good MCP connections will benefit. Companies that ignore it risk falling behind.

For the whole AI community, security is the most pressing issue to solve. MCP gives AI agents real power to do real things. That power must be matched with strong safeguards — clear rules, good tools, and careful engineering — to make sure MCP makes the world better, not more vulnerable.

Moving MCP to the Linux Foundation in December 2025 was a sign of maturity: the community recognized MCP as shared, critical infrastructure that belongs to everyone. That is the right foundation for what comes next.

References

[1] Anthropic. (2024, November 25). Introducing the Model Context Protocol. Anthropic Blog. https://anthropic.com/news/model-context-protocol

[2] Hou, X., et al. (2025). Model Context Protocol (MCP): Landscape, Security Threats, and Future Research Directions. arXiv preprint arXiv:2503.23278.

[3] Vaswani, A., et al. (2017). Attention is all you need. Advances in Neural Information Processing Systems, 30.

[4] Yao, S., et al. (2022). ReAct: Synergizing Reasoning and Acting in Language Models. arXiv preprint arXiv:2210.03629.

[5] Pento AI. (2025). A Year of MCP: From Internal Experiment to Industry Standard. https://www.pento.ai/blog/a-year-of-mcp-2025-review

[6] Andreessen Horowitz. (2025). A Deep Dive into MCP and the Future of AI Tooling. https://a16z.com/a-deep-dive-into-mcp-and-the-future-of-ai-tooling/

[7] Gupta, D. (2025). The Complete Guide to Model Context Protocol (MCP). https://guptadeepak.com/

[8] Frontegg. (2025). How MCP Will Do for AI Agents What REST APIs Did for SaaS. https://frontegg.com/blog/

[9] The Next Web. (2026, March). Rise of Model Context Protocol in the Agentic Era. https://thenextweb.com/news/rise-of-model-context-protocol-in-the-agentic-era

[10] Robison, G. (2025). The Model Context Protocol: The Architecture of Agentic Intelligence. Medium.

[11] MCP Security Working Group. (2025). MCP-scan: Automated Vulnerability Assessment for MCP Server Deployments. GitHub.